

Student Name	Student ID	Ex 1: (4 pts)	Ex 2: (4 pts)	Ex 3: (1 pt)	Ex 4: (1 pt)	Ex 5: (4 pts)	Ex 6: (4 pts)	Total (18 pts) - General Comments
alvaranalliya	u7391093	3/4 – Correct months plotted. Missing colorbar units.	2/4 – Correct region masking. Used basin-average instead of integrated rainfall; missing area calculation.	1/1 – Clear time series with labelled axes and title.	1/1 – Include interpretation of regression stats.	3.5/4 – Clear plots with thoughtful comparison. Use exact values instead of approximations (~112–122 mm) for scientific precision.	3.5/4 – Good identification of La Niña periods with seasonal analysis and IOD context. Include precise statistics rather than qualitative approximations. Use numbers.	14/18
campbellryan	u7130976	2/4 – Good progress with two figures and mm units in second attempt! Please include code for reproducibility.	0.0/4 – I see the summary numbers, but need to understand your methodology. Please show how you applied the MDB polygon.	1/1 – Time series values shown. Please include the code showing how MDB numbers are derived.	0/1 – I see the numbers, but this requires visualization and methodology — please add figures and code.	0/4 – Good narrative foundation! Please add supporting plots to demonstrate your analysis, especially for 2005–2009.	0/4 – Nice discussion of averages! Add figures to support your claims for scientific rigor.	3/18 Please submit your code — I'd love to see your approach!
dejongmilou	u8307744	4/4 – Four correct maps with BoM colormap, shared scale, and units; very clear presentation.	2/4 – MDB mask applied, but computed basin-average instead of true integrated rainfall; no area weighting.	1/1 – Time series plotted with labeled axes and units.	1/1 – Trend calculated with linear regression and result reported.	3.5/4 – Strong plots and spatial comparisons; thoughtful analysis, but some claims remain qualitative and not fully quantified.	3.5/4 – Good comparison of La Niña periods with time series and spatial maps; analysis clear, though still partly qualitative and lacking precise statistical summaries.	15/18 I like bare python scripts. But if you have a python script, you need to supply a PDF report as well for me to see the figures too.
delveslily	u7470723	4/4 – Well done.	4/4 – Correct MDB polygon and area-weighting to km^3 , well done!	1/1 – Clear MDB time-series plot with labels.	1/1 – Trend computed and shown (diff + linear fit).	3.5/4 – Strong MDB drought comparison with figures (time series + spatial) and stats (means/variance); a bit of qualitative phrasing and based on means not totals.	3.5/4 – Good La Niña comparison with plots and variance; would be stronger with precise summary stats and monthly totals.	17/18 Nice job!
dunkleyethan	u7333166	4/4 – Very creative and clear use of colormaps.	2/4 – Good use of regionmask, but stopped short of computing integrated rainfall totals.	0.5/1 – Time series shown, but unclear units and no explicit integration over area.	0.5/1 – Regression done, but based on means not true integrated values.	4/4 – Strong drought comparison with temporal and spatial averages; good analysis, but note the need to account for longitude/latitude differences in grid cell area.	4/4 – Clear La Niña comparison with plots and discussion; again, remember to weight by area when aggregating across the MDB.	15/18
easterbywoodsavannah	u7567961	4/4 Correct months plotted. The choice of colorbar is excellent as this is for precipitation.	2/4 The polygon making can be more efficient, but that is great. You forgot calculation of the surface area.	1/1 Plot looks OK. Generally try to trim the plot to the extent you want to show.	1/1 For the future try to explain why you have chosen the tools you have.	4/4 Excellent analysis and calculation of the details.	4/4 If you want to show the level the precipitation is different across time periods, play with colormap range.	4/4 Great job.
ffieldfred	u7930555	4/4 Well done! Great choice of colormap.	2/4 Nice masking approach! Please add comments explaining your surface area calculation and ensure area units are correct.	1/1 Good work! With corrected unit area, this would be excellent.	0.5/1 Close! Please correct the unit area calculation and add a visualization of the trend.	2/4 Good data analysis foundation! In scientific writing, please add plots to support your statements and conclusions.	2/4 You have the numbers — great! Please apply consistent methodology across all time periods for comparison.	10/18

gardinerbradley	u7224912	2/4 Good start with the plotting! Please add colorbar and unit labels so readers can interpret your results.	2.0/4 Nice polygon selection approach! For full marks, ensure correct area calculation for each data point.	1/1 Plot looks good! For integration analysis, please check that units match the integration requirements.	0/1 Please check your code submission — I'd like to see your approach to this problem.	0/4 Please check your code submission — looking forward to seeing your analysis.	0/4 Please check your code submission.	4/18 Please verify your submission includes all required files — happy to help if you encountered technical issues!
glovertara	u6893895	4/4 Your mm/day unit is technically correct.	3/4 Great job in calculating the surface area. But why is all values negative? You have done this amazing work, but not paid attention to the units.	1/1	1/1	1/4 It is funny how you have the title 2005-2009 for the first plot, but the x axis is showing 2016→. Good thinking about some details. Sadly not paying attention to a few others. Also, the reason all values are negative is not for being below average, it is the way you have computed surface area being negative.	1/4 I think you are missing the point that all values are negative! You interpret as too negative as being too little, which I think is the other way around.	11/18 You are clearly good in the way you think about problems! But somehow you have managed everytime to ignore the outstanding issues.
kashkarovstephan	u6729293	0/4 you are required to plot the monthly values geographically (if not using cartopy and more)	2/4 No calculation of unit area to compute total volume. The hand drawn markers look odd for a computer lab assignment. But I appreciate the analysis.	1/1	1/1	4/4	4/4	12/18
laibutlersaskia	u7943537	4/4 Nice choice of colormap.	4/4 Well done! One of the few accounting for surface area. You could have just stayed with the total volume.	1/1 Correct	4/4 Correct.	4/4	4/4	18/18
lemmjosie	u7932320	4/4 Nice arrangement of the plots.	2/4 That does not account for the difference in surface area.	1/1	1/1	4/4	4/4	16/18
mannjustin	u8350605	4/4 Nice! Technically true about mm/month (but it's just the average of precipitation in that month)	2/4 Good job showing how you figured out working out with the polygons. But you did not account for the surface area.	1/1	1/1 I did not get what you wanted to do with the external data set	4/4 You do here account for the surface area by just introducing the total surface of the MDB. You should be able to calculate that having the coordinates of the polygon.	4/4	16/18

mcfaddenben	u7655069	4/4 Nice.	2/4 Good approach! For full integration, please account for surface area to calculate the total volume of rain rather than just averaging.	1/1	1/1 Correct. Interesting to see someone using <code>statsmodels.api</code> !	3/4 Good foundation! Please strengthen your analysis by adding specific quantitative comparisons between the 2005-2009 and 2016-2019 periods with actual numbers.	3/4 You have good code — please add more detailed quantitative analysis to interpret your results.	14/18
meixnersarah	u7930427	3/4 No units for the plot.	2/4 You have not accounted for the surface area for each point	1/1 OK.	0/1 Not showing any code here, it's very hard to know what you have done. The values seem consisten with other students.	2/4 You have not looked a the time-series by plotting it and looking at the patterns, instead you have only looked at the mean(I suppose? There is no information on that!) But you have done enough research to get some points for this.	2/4 As before	10/18
pascoedecan	u7478492	1/4 Good attempt at the four-panel layout! Please add colorbar and units for scientific interpretation.	4/4 Excellent! One of the few who properly accounted for surface area — well done!	1/1	1/1 Good approach! For future assignments, consider adding more statistical analysis (rates, distributions) for deeper insights.	0/4 Great visual approach! Please add units to your maps for scientific rigor. Remember, this assignment tests your ability to analyze the provided data directly.	0/4 Please apply the same scientific analysis standards as above.	7/18 Strong work on surface area calculation! Focus on adding units and direct data analysis to strengthen your scientific presentation.
rechnerthomasmax	u7468592	4/4 Great!	4/4 Very good! Best I have seen in dealing with surface area.	1/1	0.5/1 I see how you thought this could be the answer. But generally, and specially since this is data cross decades, there is so much more to data that might get lost if you focus on the starting and end points. Something like fitting a line would be better.	4/4 Bang on! Very good job.	4/4 Great	17/18
shankarsheryl	u7143617	4/4 Nice!	2/4 Good polygon masking approach! For “integrated rainfall,” please add surface area calculations to convert your averages to true integrals.	1/1	1/1 Very good.	2/4 Great plotting work! Now please add quantitative analysis to interpret your visual results — the numbers will strengthen your conclusions.	1/4 Please add more quantitative explanation to support your visual analysis.	11/18
simpsonsophie	u7304948	4/4 Nice! Excellent work having the same colorbar for all panels — great for consistent comparison!	2/4 Good foundation! Please add surface area calculations for true spatial integration.	1/1	1/1 Technically correct. Consider adding more visualization for future assignments!	2/4 Good external research! Please add direct quantitative comparison using the data you've plotted and calculated.	2/4 Please add more quantitative analysis to support your conclusions with the data you've generated.	12/18
taibmaya	u7646490	4/4 Excellent plotting.	2/4 Excellent everything! But we were interested in the integrated rainfall! So we need to account for the surface area somehow. Either by approximation, or estimating the surface associated with each point.	1/1	1/1 Great.	4/4	4/4 Exemplary analysis.	17/18

tapsfieldcharlie	u7655929	4/4 Well done!	2/4 Good approach! For “integrated rainfall,” please add surface area calculations for each point.	1/1	1/1 Interesting to see someone using polyfit.	2/4 Please strengthen your analysis with more quantitative and scientific methods.	2/4 Please add more detailed quantitative analysis to support your conclusions.	12/18
wilsonciara	u7471093	4/4 Well done for having the same colorbar.	2/4 Nice job! But since we are interested in the integrated rainfall, we need to account for the surface area.	1/1	1/1 Interesting to see someone using polyfit.	4/4 A generous full mark. But next time use more quantitative values like you have already. But scientific analysis is all about that.	4/4 Well done! Specially for doing the difference plot.	16/ 18
yakemichaeljames	u8301677	4/4 Excellent plots.	2/4 Interesting to see lines 15-22. Was this really necessary? Why do you need to test all these different cases? You also need to account for the surface area of each point.	1/1	1/1	4/4 Very nice and unique way of analysing data by using the surface area, for basing time integration.	4/4	16/18
yancrystal	u7894912	4/4	2/4 Good foundation! Please add surface area calculations for full integration.	1/1	1/1	0/4 Please provide your code for reproducibility and verification.	0/4 Please include code to support your statistical analysis. Some references appear to need verification.	8/18 Please submit your code to demonstrate your methodology!
yousufzaiahmadsurosh	u7513423	4/4 Nice job. Units and single colorbar for all figures	4/4 Great job in calculating the surface area.	1/1	1/1 Correct.	2/4 Good foundation! Please expand your pattern analysis beyond total sums — consider variability, frequency, and temporal patterns.	2/4 Please add more detailed analysis as suggested above.	14/18

Table 0.1

0.1 Exercise Descriptions

- **Ex 1 (4 pts):** Plot actual rainfall for Jan 2005, Aug 2009, Feb 2010, Dec 2019 as four subplots
- **Ex 2 (4 pts):** Derive integrated rainfall for each month within Murray-Darling Basin
- **Ex 3 (1 pt):** Plot time series of monthly rainfall integrated over Murray-Darling Basin
- **Ex 4 (1 pt):** Calculate rate of change in monthly rainfall over time span
- **Ex 5 (4 pts):** Compare drought periods 2005–2009 vs 2016–2019
- **Ex 6 (4 pts):** Compare La Niña periods 2010–2012 vs 2020–2023

Total: 18 points